2. Non-Functional Requirements

Usability

Responsive interface that adapts to screen size.

Reliability

The system must operate 24/7 with a reliability of > 99.5%.

Performance

The system must handle 2,000 concurrent users with zero errors.

The system must scale to handle over 100,000 customers, 10,000 vehicles, and 1,000,000

bookings without degrading response times

Supportability

None

Security

Customers can only view their own information and orders. Data must be encrypted at rest and

viewable only by authorized staff. Use parameterized queries (SQL injection prevention),

output encoding (XSS prevention), and CSRF tokens on all state-changing requests.

Design Constraints

HTML5, CSS3, JavaScript front end; server-side language of the team's choice and MySQL

database management system

3. Data Flow Diagrams

3.1 DFD Level 0 (context diagram)

The whole system is one process. It exchanges data with three external entities: the Customer, the Admin/staff and the Payment gateway.

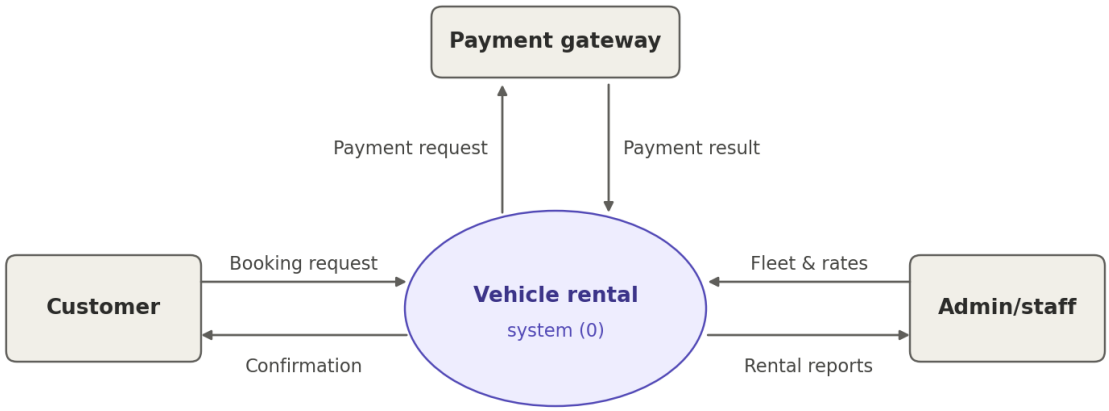


Figure 1. DFD Level 0 – Vehicle Rental System

Data flow	From	To	Data carried
Booking request	Customer	System (0)	Registration and login details, search criteria, pickup and return dates, vehicle choice, payment choice, return of the vehicle
Confirmation	System (0)	Customer	Search results, booking confirmation, invoice, rental status, email notice
Fleet & rates	Admin/staff	System (0)	Vehicle details, daily rates, fees, rental handover and return inspection, account administration
Rental reports	System (0)	Admin/staff	Booking lists, utilisation and revenue reports, vehicle status
Payment request	System (0)	Payment gateway	Amount, booking reference, customer payment redirect
Payment result	Payment gateway	System (0)	Success or failure, transaction reference

3.2 DFD Level 1

Process 0 is split into five sub-processes and three data stores. Arrows are double-headed because each link carries a request in one direction and a response in the other.

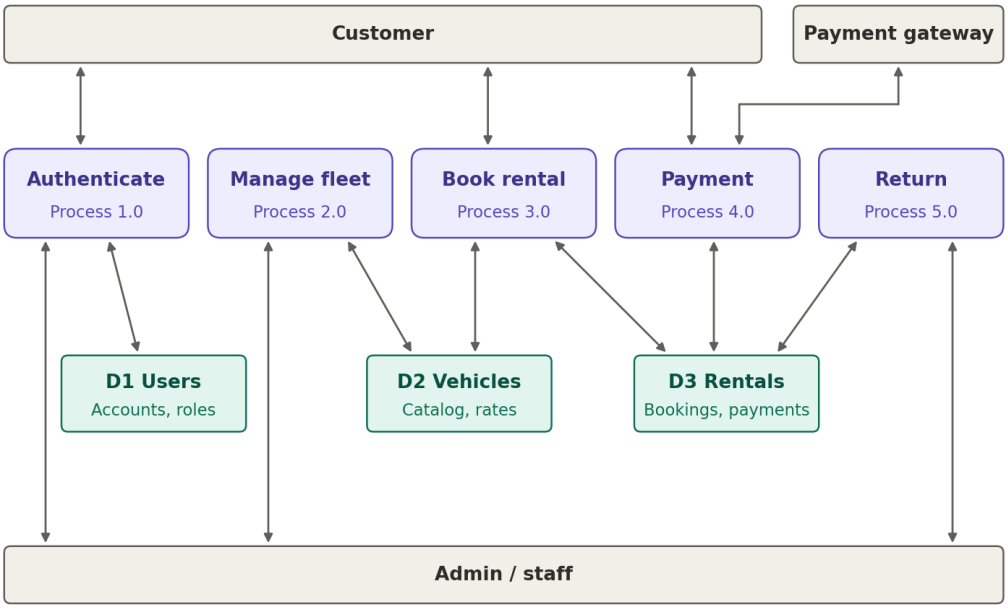


Figure 2. DFD Level 1 – processes, data stores and external entities

Process	Name	Inputs	Outputs	Data stores
1.0	Authenticate	Registration details and login credentials from Customer and Admin/staff	Login result, session, role	D1 Users (read/write)
2.0	Manage fleet	Vehicle details, rates and status changes from Admin/staff	Updated vehicle list and confirmation	D2 Vehicles (read/write)
3.0	Book rental	Search criteria, dates and vehicle choice from Customer; vehicle catalog from D2	Search results, booking record, confirmation; availability checked against dates already booked in D3	D2 (read), D3 (read/write)
4.0	Payment	Payment choice from Customer; booking total from D3; result from Payment gateway	Payment request to gateway, invoice and receipt to Customer, payment status	D3 (read/write)
5.0	Return	Return and inspection details from Admin/staff	Final charges, rental status 'closed', return report	D3 (read/write)

3.3 Consistency and assumptions

- Customer, Admin/staff and Payment gateway appear in both Level 0 and Level 1.
- The Customer flows of Level 0 are covered by processes 1.0, 3.0 and 4.0; the Admin/staff flows by 1.0, 2.0 and 5.0; the Payment gateway flows by 4.0 only.
- Confirmation emails are sent inside processes 3.0 and 4.0, so no separate email entity is drawn.
- D3 Rentals holds both bookings and payments to keep the diagram simple. In the database design they can be split into two tables.
- Vehicle availability is derived from the dates already booked in D3, so process 5.0 does not need to write to D2.